Cyclic substitutions and a spectral short-cycle conjecture

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Abstract

The spectral short-cycle statement attributed to Charbit in Sullivan's survey asserts that an n-vertex loopless digraph with adjacency spectral radius at least n/k has a directed cycle of length at most k. We show that this statement holds for positive integers k exactly when $k \leq 2$. For the negative cases, we apply a previously studied construction obtained by substituting transitive tournaments into a directed cycle. An explicit positive eigenvector gives its spectral radius and yields counterexamples for every $k \geq 3$. The cases $k = 3, 4, 5$ have witnesses on twenty, fifteen and twelve vertices, respectively. The proofs are self-contained apart from standard Perron–Frobenius facts. A companion Lean formalization verifies the twelve-vertex counterexample, including its strict spectral inequality.

1 The statement and the counterexamples

For a finite simple loopless digraph D on $n \geq 1$ vertices, let $A = A(D)$ be its zero–one adjacency matrix, with $A_{uv} = 1$ precisely when $u \to v$. Write $g(D)$ for its directed girth, with $g(D) = \infty$ if D is acyclic, and $\rho(A)$ for the maximum modulus of its complex eigenvalues. For positive integer k, the statement recorded as Conjecture 6.18 of Sullivan [5], attributed there to Charbit, is

$$\rho(A) \geq \frac{n}{k} \quad \Longrightarrow \quad g(D) \leq k. \tag{1}$$

We address this formulation exactly as printed. In particular, it imposes no regularity or Eulerian hypothesis. An earlier spectral formulation appears in Charbit's thesis [3, Section 5.4].

The construction used below is already present in the literature. Chudnovsky, Seymour and Sullivan [4, Section 1] give $C_4[T_s]$ as an extremal example for a feedback-arc-set problem. Aharoni, Berger, Chudnovsky, Guo and Zerbib [1, Section 2] describe the general family $C_g[T_s]$, crediting Hompe's 2019 thesis, and use it to disprove a proposed upper bound on directed girth in terms of reciprocal outdegrees. We apply this established family to (1) by evaluating its adjacency spectral radius. The graph family and the substitution operation are not claimed as new.

Theorem 1. *For a positive integer k, implication* (1) *holds for all finite simple loopless digraphs on at least one vertex if and only if $k \leq 2$. For every $k \geq 3$ there is a simple, strongly connected oriented graph D such that $g(D) = k + 1$ and $\rho(A(D)) > |V(D)|/k$.*

2 The two positive cases

For $k = 1$, every row sum is at most $n - 1$, so $\rho(A) \leq n - 1 < n$ and the antecedent is impossible.

For $k = 2$, suppose that D has no digon. Then $A + A^{\mathsf{T}} \leq J - I$ entrywise, where J is the all-ones matrix. A nonnegative matrix, including a reducible one, has a nonzero nonnegative real vector x

with $Ax = \rho(A)x$; see, for example, Butler–Siegel [2, Remark 3.1]. As $x^{\mathsf{T}}x > 0$, Cauchy–Schwarz gives

$$2\rho(A)x^{\mathsf{T}}x = x^{\mathsf{T}}(A + A^{\mathsf{T}})x \le \left(\sum_i x_i\right)^2 - x^{\mathsf{T}}x \le (n-1)x^{\mathsf{T}}x.$$

Thus $\rho(A) \le (n-1)/2 < n/2$. These standard arguments establish the positive part of Theorem 1, including $n = 1$.

3 Cyclic substitution

Let $g \ge 3$ and $s \ge 1$ be integers. Define $D_{g,s}$ on

$$V(D_{g,s}) = \mathbb{Z}/g\mathbb{Z} \times \{0, \dots, s-1\}$$

by putting in exactly the arcs

$$(i,a) \to (i,b) \quad (a < b), \qquad (i,a) \to (i+1,b) \quad (0 \le a, b < s). \tag{2}$$

Equivalently, $D_{g,s} = C_g[T_s]$, with T_s the transitive tournament of order s. We use the construction described in [4, 1].

Lemma 2. *The digraph $D_{g,s}$ is simple, oriented and strongly connected. Moreover,*

$$|V(D_{g,s})| = gs, \qquad |E(D_{g,s})| = g\left(s^2 + \binom{s}{2}\right), \qquad g(D_{g,s}) = g,$$

and

$$\rho(A(D_{g,s})) = \frac{1}{2^{1/s} - 1}. \tag{3}$$

Proof. The counts follow from (2). There are no loops or opposite arcs, since $g \ge 3$ and the arcs within a class follow a strict order. Crossing arcs allow every vertex to reach every other vertex, going once around the classes if necessary.

A directed cycle cannot use only within-class arcs. The positive number of crossing arcs in a cycle is divisible by g, since each crossing adds one to the class index modulo g. Its length is therefore at least g. Keeping the second coordinate zero gives a cycle of length g.

Put $r = 2^{1/s} > 1$, $\lambda = (r-1)^{-1}$, and $x(i,a) = r^{s-1-a}$. Since $r^s = 2$,

$$S := \sum_{b=0}^{s-1} r^{s-1-b} = \frac{r^s - 1}{r - 1} = \lambda.$$

The sum of the coordinates on out-neighbors of (i,a) is

$$(Ax)(i,a) = S + \sum_{b=a+1}^{s-1} r^{s-1-b} = \lambda + \frac{r^{s-1-a} - 1}{r-1} = \lambda x(i,a).$$

Thus λ is an eigenvalue with a strictly positive eigenvector. For completeness, let $Az = \mu z$ with $z \ne 0$ complex, and choose u maximizing $|z_u|/x_u$. This maximum is positive, and

$$|\mu|\,|z_u| \le \sum_v A_{uv}|z_v| \le \frac{|z_u|}{x_u}\sum_v A_{uv}x_v = \lambda|z_u|.$$

Hence every eigenvalue satisfies $|\mu| \le \lambda$, which proves (3). □

Proof of the negative part of Theorem 1. Use $D_{k+1,5}$. It has $n = 5(k+1)$ vertices and directed girth $k+1$. The integer inequality

$$23^5 = 6\,436\,343 > 6\,400\,000 = 2 \cdot 20^5$$

gives $2^{1/5} < 23/20$ and hence

$$\rho(A) = \frac{1}{2^{1/5} - 1} > \frac{20}{3} \geq \frac{5(k+1)}{k} = \frac{n}{k},$$

where the second inequality uses $k \geq 3$. □

4 Small witnesses and exact certificates

Smaller tournament sizes suffice at $k = 4$ and at every $k \geq 5$. For $k = 4$, use $D_{5,3}$: the inequality $19^3 > 2 \cdot 15^3$ gives $\rho(A) > 15/4$. For $k \geq 5$, use $D_{k+1,2}$, since

$$\rho(A) = 1 + \sqrt{2} > \frac{12}{5} \geq \frac{2(k+1)}{k}.$$

Thus three small instances are as follows; their orders are not claimed to be minimal.

k	Digraph	Vertices	Arcs	Girth	Spectral radius
3	$D_{4,5}$	20	140	4	$(2^{1/5} - 1)^{-1} > 20/3$
4	$D_{5,3}$	15	60	5	$(2^{1/3} - 1)^{-1} > 15/4$
5	$D_{6,2}$	12	30	6	$1 + \sqrt{2} > 12/5$

Short integer certificates give another proof of every negative case. Assign the same positive vector w to each class. Direct sums of outgoing weights give the following vectors and residuals, displayed per class:

$$\begin{aligned}
&k = 3: \quad w = (33, 29, 25, 22, 19), \quad 3Aw - 20w = (9, 2, 7, 1, 4),\\
&k = 4: \quad w = (11, 9, 7), \quad 4Aw - 15w = (7, 1, 3),\\
&k \geq 5: \quad w = (17, 12), \quad kAw - 2(k+1)w = (7k - 34, 5k - 24).
\end{aligned}$$

In the last row, the residuals equal $7(k-5)+1$ and $5(k-5)+1$, so every residual is strictly positive. The standard positive-vector bound gives $\rho(A) > n/k$; see [2, Theorem 1.2]. For completeness, if $y \geq 0$ is a nonzero left Perron vector, then $y^{\mathsf{T}} w > 0$, and $Aw > cw$ gives

$$\rho(A) y^{\mathsf{T}} w = y^{\mathsf{T}} A w > c\, y^{\mathsf{T}} w.$$

This also applies to reducible nonnegative matrices. These integer certificates supplement the closed-form proof above.

The minimum outdegree of $D_{g,s}$ is s, attained at $(i, s-1)$. In our examples $g = k+1$, so $s < gs/k$. Consequently they do not satisfy the degree hypothesis in the ordinary Caccetta–Häggkvist conjecture.

5 Formal verification of the twelve-vertex example

The twelve-vertex example also has a Lean 4 formalization, maintained with the companion sources at

https://github.com/alexgocardinal/charbit-spectral-counterexamples.

The development uses Lean 4.24.0 and Mathlib at tag `v4.24.0`. In the source version `ee815b3`, the theorem `Charbit.counterexample` verifies looplessness, the absence of opposite arcs, the absence of nonempty closed directed walks of lengths one through five, and the strict inequality $\rho(A) > 12/5$. Here the spectral radius is Mathlib's standard supremum of the moduli of the complex spectrum, applied to the complexified zero–one adjacency matrix. The proof verifies the eigenvalue $1 + \sqrt{2}$ using the repeated vector $(\sqrt{2}, 1)$ and then bounds the spectral radius from below by its modulus. The arcs of the displayed six-cycle are checked separately.[1]

This formalization covers the twelve-vertex counterexample. The variable-order family and the classification in Theorem 1 are proved in the preceding sections and are not asserted to have been formalized.

References

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[5] B. D. Sullivan, *A Summary of Results and Problems Related to the Caccetta–Häggkvist Conjecture*, AIM 2006–13, arXiv:math/0605646v1, 2006, Conjecture 6.18. https://arxiv.org/abs/math/0605646.

[1] The source files are `Charbit.lean` and `Verify.lean`. The commands `lake build` and `lake env lean Verify.lean` reproduce the build and assumption audit. The counterexample theorem depends only on Lean's standard foundational axioms `propext`, `Classical.choice` and `Quot.sound`; there are no admitted proof steps. The repository is private during manuscript preparation.